

Smart Helmet Safety Reminder and Guardian Monitoring System

TN Skills | VTPT (Vetri Thiran Payirchi Thittam) | Individual Hackathon

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The problem

In 2022, **50,029 people** died in Indian road accidents **without a helmet**, nearly 3 in 10 of all road deaths (MoRTH, Road Accidents in India 2022). Riders **forget**, skip it for a short ride, and existing reminders talk to the **rider only**. The family has **no visibility** of the habit.

My solution

A low-cost **ESP32 IoT system** that works like a bike's side-stand warning, but for the helmet. Before the ride, a **red light and buzzer** warn the rider if the registered helmet is not found or not worn; when it is, a **green light** shows safe and the prototype permits the operation signal. Every trip is recorded, so a parent or guardian sees the **daily, weekly and monthly pattern** and gets a **GSM SMS** after repeated misses.

How it works

1 Detect ESP32 finds the helmet's unique ID over Bluetooth	2 Verify Wear sensor confirms it is on the head	3 Decide Ignition: green + permit if safe, red + buzzer if not	4 Record Trip logged to ThingSpeak and Google Sheet	5 Alert 3 misses in a row: SMS to the guardian
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Only four tools, chosen on purpose

Wokwi	Simulate the ESP32, switches, LEDs and buzzer
ThingSpeak	Live status and graphs from the ESP32
Google Sheet + Apps Script	Trip log, helmet register, guardian email alert, dashboard data
Web dashboard	Guardian view: live status, trends, alerts

Real-build hardware: ESP32 (Bluetooth + Wi-Fi), BLE tag in the helmet, chin-strap hall or IR wear sensor, LEDs and buzzer, SIM800L GSM module. In Wokwi, a slide switch stands in for Bluetooth detection, because Wokwi cannot simulate BLE.

Simulation and results

Screenshots and links of the working prototype. Paste each image into its box and the link below it.

[Paste screenshot here] Wokwi simulation	[Paste screenshot here] ThingSpeak graphs
Wokwi simulation link: [paste link here]	ThingSpeak graphs link: [paste link here]
[Paste screenshot here] Google Sheet log	[Paste screenshot here] Guardian dashboard
Google Sheet log link: [paste link here]	Guardian dashboard link: [paste link here]

Test cases shown

Safe ride: helmet detected and worn, green and permit LED on, trip logged as OK. **Not worn / not detected:** red LED and buzzer, trip logged as violation. **Three misses in a row:** guardian alert (SMS line in the serial monitor, email from Apps Script).

Future scope

Helmet-lock detection, replacing a registered helmet from the dashboard, ESP32-CAM secondary verification, and GPS to find a misplaced helmet.

Conclusion

The goal is not only to remind the rider, but to create a **continuous helmet-usage link** between the rider and a trusted guardian. A helmet cannot save anyone from the shelf, so this prototype tries to make wearing it the **default**.